Program 3

Define a structure called Time containing 3 integer members (hour, minute, second). Develop a menu driven program to perform the following by writing separate function for each operation. a) Read (T): To read time b) Display (T):To display time c) update(T):To Update time d) Add (T1, T2) : Add two times. Update function increments the time by one second and returns the new time (if the increment results in 60 seconds, then the second member is set to zero and minute member is incremented by one. If the result is 60 minutes, the minute member is set to zero and the hour member is incremented by one. Finally, when the hour becomes 24, Time should be reset to zero. While adding two time variable, normalize the resultant time value as in the case of update function.
Note: Illustrate the use of pointer to pass time variable to different functions.

```
#include <stdio.h>

#include <stdlib.h>

// Define the Time structure

typedef struct Time {

int hour;

int minute;

int second;

} Time;

// Function to normalize time (handle overflows for seconds, minutes, hours)

void normalizeTime(Time *t) {

t->minute += t->second / 60;

t->second %= 60;

t->hour += t->minute / 60;

t->minute %= 60;

t->hour %= 24; // Reset hour to 0 if it reaches 24

}

// a) Read Time

void readTime(Time *t) {

printf("Enter hour: ");

scanf("%d", &t->hour);
```

```
printf("Enter minute: ");
scanf("%d", &t->minute);
printf("Enter second: ");
scanf("%d", &t->second);
normalizeTime(t); // Normalize immediately after reading
}
// b) Display Time
void displayTime(const Time *t) {
printf("%02d:%02d:%02d\n", t->hour, t->minute, t->second);
}
// c) Update Time (increment by one second)
void updateTime(Time *t) {
t->second++;
normalizeTime(t);
}
// d) Add two times
Time addTimes(const Time *t1, const Time *t2) {
Time result;
result.second = t1->second + t2->second;
result.minute = t1->minute + t2->minute;
result.hour = t1->hour + t2->hour;
normalizeTime(&result);
return result;
}
int main() {
Time t1, t2, t_sum;
int choice;
do {
```

```
printf("\n--- Time Operations Menu ---\n");
printf("1. Read Time (T1)\n");
printf("2. Display Time (T1)\n");
printf("3. Update Time (T1)\n");
printf("4. Read Time (T2)\n");
printf("5. Display Time (T2)\n");
printf("6. Update Time (T2)\n");
printf("7. Add Two Times (T1 + T2)\n");
printf("0. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
printf("Reading Time T1:\n");
readTime(&t1);
break;
case 2:
printf("Displaying Time T1: ");
displayTime(&t1);
break;
case 3:
printf("Updating Time T1 by one second...\n");
updateTime(&t1);
printf("New Time T1: ");
displayTime(&t1);
break;
case 4:
printf("Reading Time T2:\n");
```

```
readTime(&t2);
break;
case 5:
printf("Displaying Time T2: ");
displayTime(&t2);
break;
case 6:
printf("Updating Time T2 by one second...\n");
updateTime(&t2);
printf("New Time T2: ");
displayTime(&t2);
break;
case 7:
printf("Adding T1 and T2...\n");
t_sum = addTimes(&t1, &t2);
printf("Sum of times: ");
displayTime(&t_sum);
break;
case 0:
printf("Exiting program.\n");
break;
default:
printf("Invalid choice. Please try again.\n");
}
} while (choice != 0);
return 0;
}
```